CUSTOMER CHURN PREDICTION IN TELECOM USING ML

A machine learning approach to predict and prevent customer churn in telecommunications

GROUP-20

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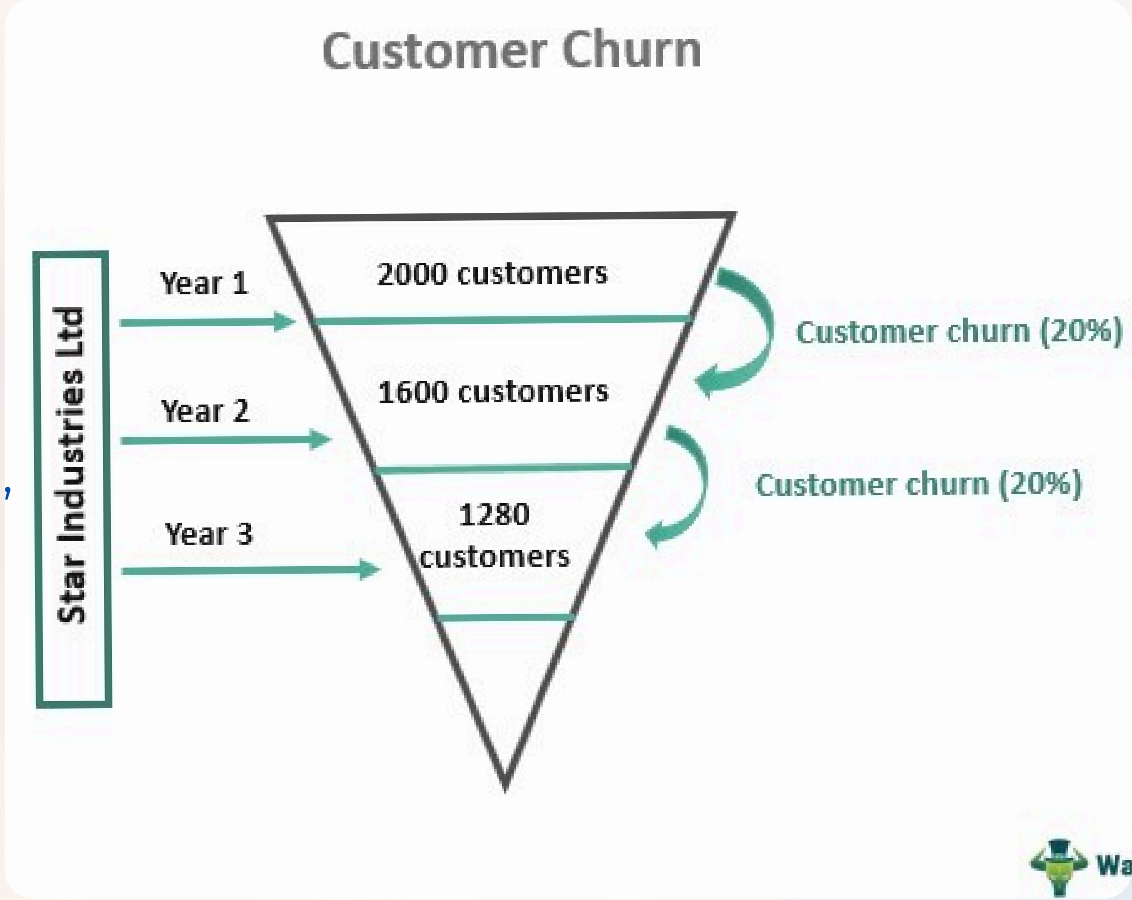
UNDERSTANDING CUSTOMER CHURN

- Customer churn occurs when subscribers discontinue their telecom services and switch to competitors.
- This phenomenon represents a critical business challenge that directly impacts revenue streams and market position. In the highly competitive telecom landscape, acquiring new customers costs 5-7 times more than retaining existing ones.
- Understanding churn patterns enables proactive intervention and strategic retention efforts.



PROBLEM STATEMENT & STRATEGIC OBJECTIVE

- CUSTOMER DATA
- MACHINE LEARNING MODEL
- PREDICT CHURN
- BUSINESS RETENTION STRATEGY
- REDUCED REVENUE LOSS

- Preventing customer churn is not just about retaining clients; it directly safeguards a telecom company's recurring revenue stream.
- Since acquiring new customers is significantly more expensive than retaining current ones, focusing on churn reduction helps companies optimize their marketing and operational costs.
- By understanding the reasons customers leave, telecom providers can refine their services and build loyalty programs that drive long-term customer satisfaction and competitive advantage.

DATA COLLECTION

- We used the Telco Customer Churn dataset, which contains 7043 customers and 21 features.
- These include customer demographics (gender, senior citizen, dependents), service details (phone, internet, streaming), billing information (monthly charges, total charges), and contract type.
- The target variable is Churn (Yes/No), which indicates whether a customer left the service or stayed.

```python
import pandas as pd
df = pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
print(df.shape)
print(df.columns)
print(df.info())
print(df['OnlineSecurity'].unique())
print(df['TechSupport'].unique())
print(df['Partner'].unique())
print(df['Dependents'].unique())
print(df['PhoneService'].unique())
print(df['MultipleLines'].unique())
print(df['OnlineBackup'].unique())
print(df['DeviceProtection'].unique())
print(df['StreamingTV'].unique())
print(df['StreamingMovies'].unique())
print(df['PaperlessBilling'].unique())
print(df['Churn'].unique())
```

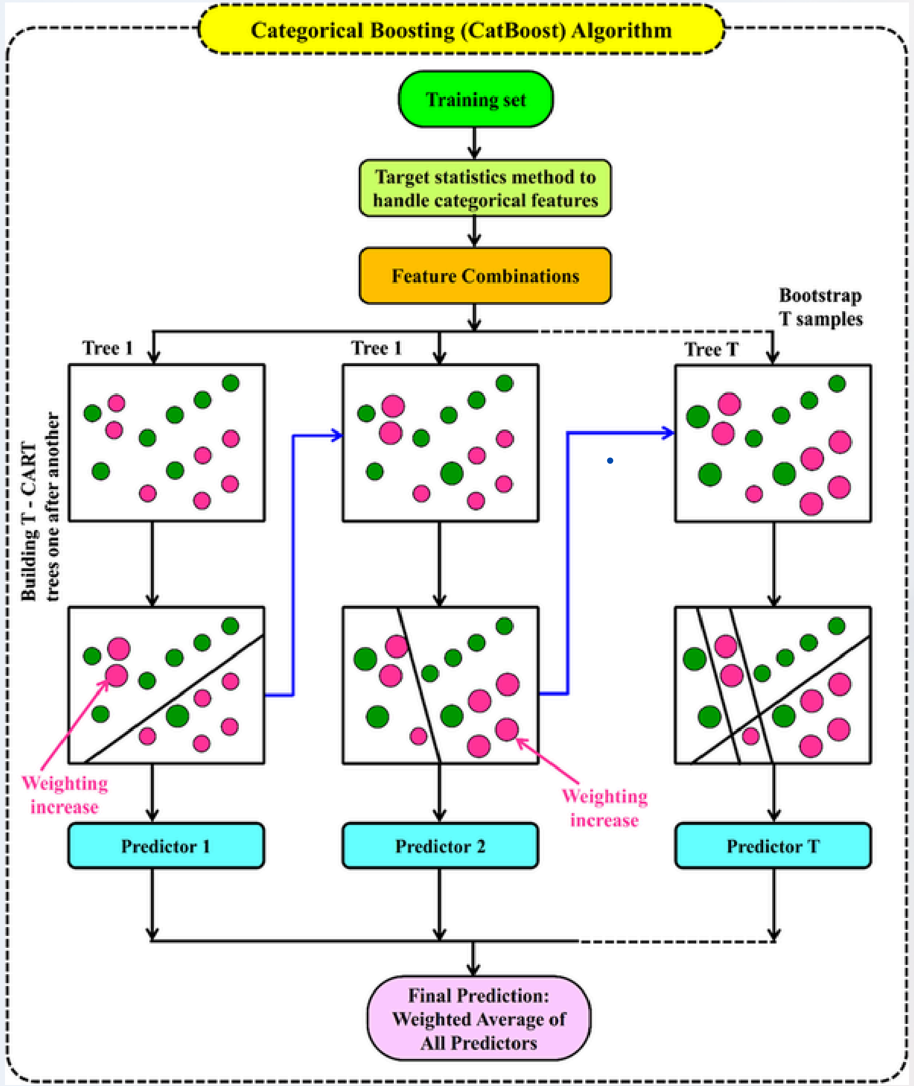
```
(7043, 21)
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
```

DATA PRE-PROCESSING

```python
# ====== IMPORTS ======
import pandas as pd
import matplotlib.pyplot as plt
from catboost import CatBoostClassifier, Pool
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# ====== LOAD DATA ======
df = pd.read_csv('/content/WA_Fn-UseC_-Telco-Customer-Churn.csv')

# ====== PREPROCESSING ======
# TotalCharges to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'].replace(" ",0))

# Replace 'No internet service' or 'No phone service' with 'No'
replace_cols = ['MultipleLines','OnlineSecurity','OnlineBackup','DeviceProtection',
                'TechSupport','StreamingTV','StreamingMovies']
for col in replace_cols:
    df[col] = df[col].replace({'No internet service':'No','No phone service':'No'})

# Binary mapping
binary_cols = ['Partner','Dependents','PhoneService','PaperlessBilling','Churn']
for col in binary_cols:
    df[col] = df[col].map({'Yes':1,'No':0})

# Feature engineering
df['NumServices'] = ((df['PhoneService']==1).astype(int) +
                     (df['MultipleLines']=='Yes').astype(int) +
                     (df['OnlineSecurity']=='Yes').astype(int) +
                     (df['OnlineBackup']=='Yes').astype(int) +
                     (df['DeviceProtection']=='Yes').astype(int) +
                     (df['TechSupport']=='Yes').astype(int) +
                     (df['StreamingTV']=='Yes').astype(int) +
                     (df['StreamingMovies']=='Yes').astype(int))


# Tenure group
df['TenureGroup'] = pd.cut(df['tenure'], bins=[0,12,24,48,60,72], labels=[1,2,3,4,5])
df['TenureGroup'] = df['TenureGroup'].astype(str)

# CatBoost categorical features
cat_features = ['gender','MultipleLines','InternetService','OnlineSecurity','OnlineBackup',
                'DeviceProtection','TechSupport','StreamingTV','StreamingMovies',
                'Contract','PaymentMethod','TenureGroup']
```

- The raw dataset contained some inconsistencies which were cleaned during preprocessing.
- We converted Total Charges into numeric format and replaced blank values with 0. Categorical features like “No internet service” and “No phone service” were simplified to “No.”
- Binary columns such as Partner, Dependents, and PaperlessBilling were mapped as Yes=1 and No=0.
- We also created new features: NumServices (total number of subscribed services) and TenureGroup (grouping tenure into ranges). This ensured the data was ready for training the model.

WHAT IS CATBOOST?



- Catboost is a gradient boosting algorithm on decision trees, developed by yandex.
- It can handle categorical features automatically without manual encoding
- It uses ordered boosting to avoid overfitting and target leakage
- It gives high accuracy with minimal tuning and also supports GPU training

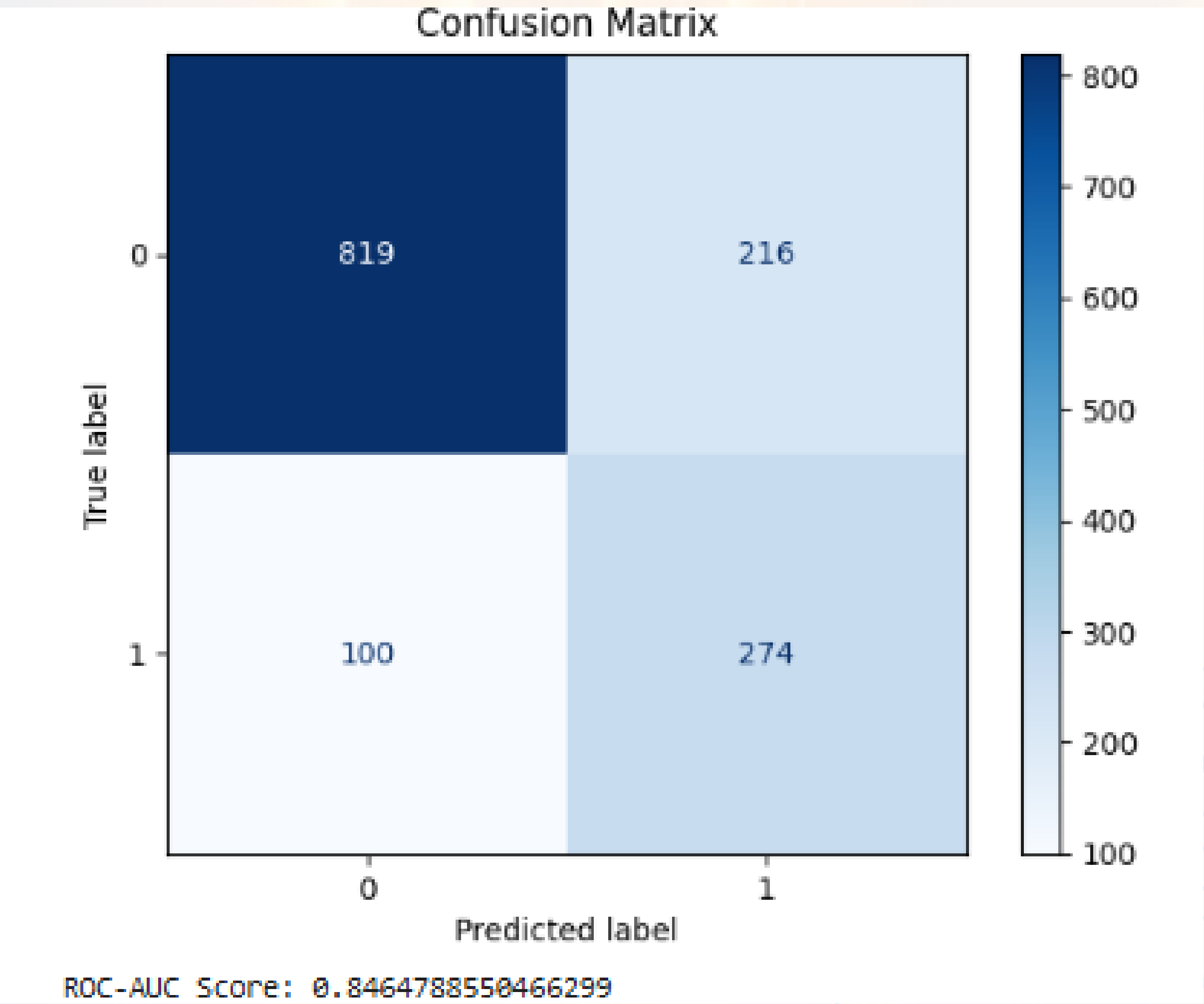
MODEL DEVELOPMENT AND IMPLEMENTATION

- We split the data into 80% training and 20% testing sets.
- We selected CatBoostClassifier as our model because it handles categorical features efficiently and provides high accuracy.
- The model was trained with 1000 iterations, depth of 6, and learning rate of 0.1. To handle class imbalance (more non-churners than churners), class weights were applied.
- The model was trained using customer demographics, service usage, billing patterns, and tenure information to predict churn.

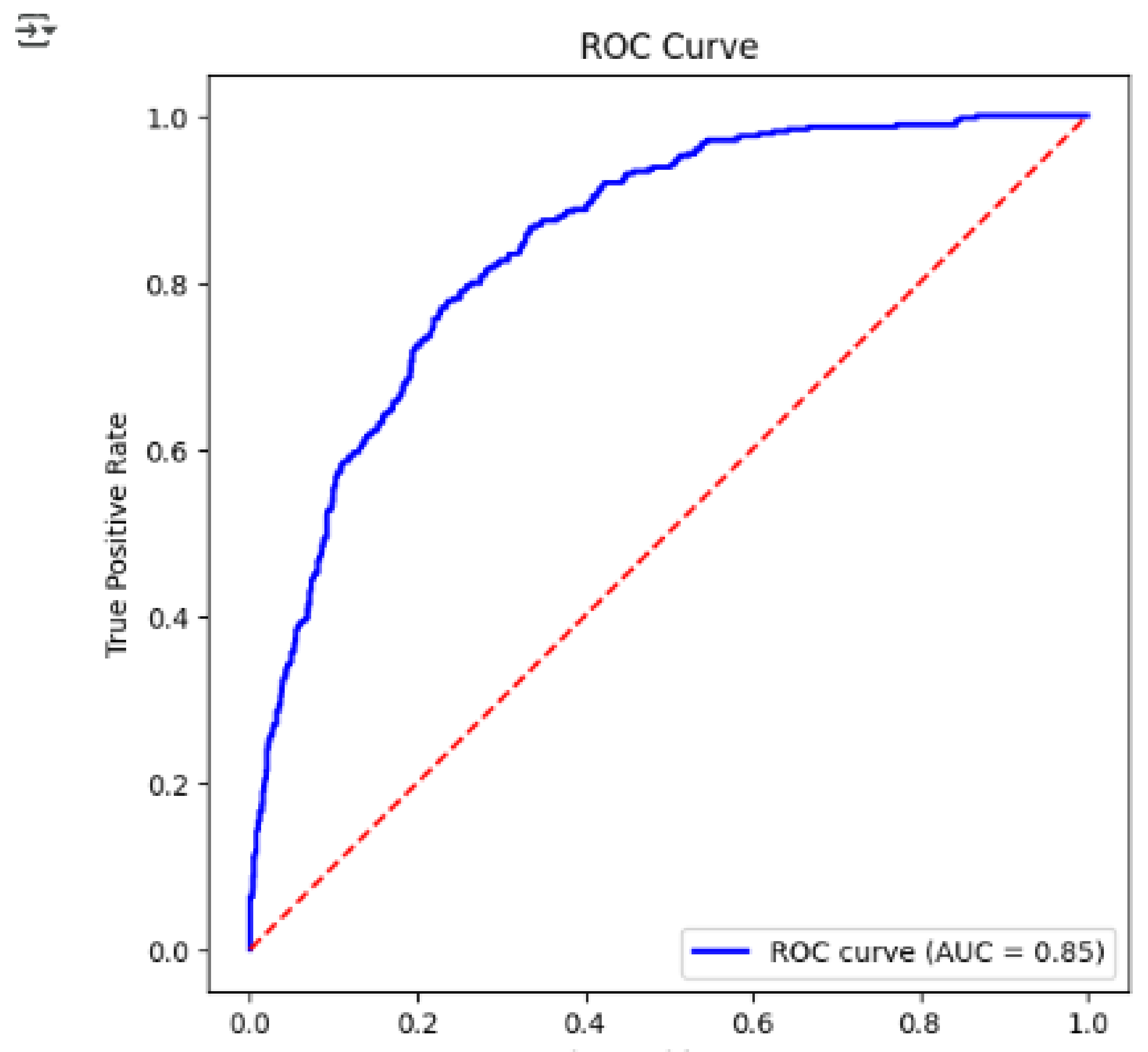
```python
# ====== TRAIN-TEST SPLIT ======
X = df.drop(['customerID','Churn'], axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42
)

train_pool = Pool(data=X_train, label=y_train, cat_features=cat_feature
test_pool = Pool(data=X_test, label=y_test, cat_features=cat_features)

# ====== TRAIN CATBOOST ======
cb_model = CatBoostClassifier(
    iterations=1000,
    depth=6,
    learning_rate=0.1,
    class_weights=[1,2],  # Handle class imbalance
    random_seed=42,
    verbose=100
)
cb_model.fit(train_pool, eval_set=test_pool)
```

PERFORMANCE EVALUATION RESULTS

- The trained model was evaluated using standard metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- The confusion matrix helped us understand how many customers were correctly predicted as churners or non-churners.
- Feature importance analysis showed that tenure, contract type, and monthly charges were the most influential factors in predicting churn.
- Overall, the model achieved strong performance, making it suitable for business use.



ROC- AUC CURVE



INTERPRETATION OF RESULTS

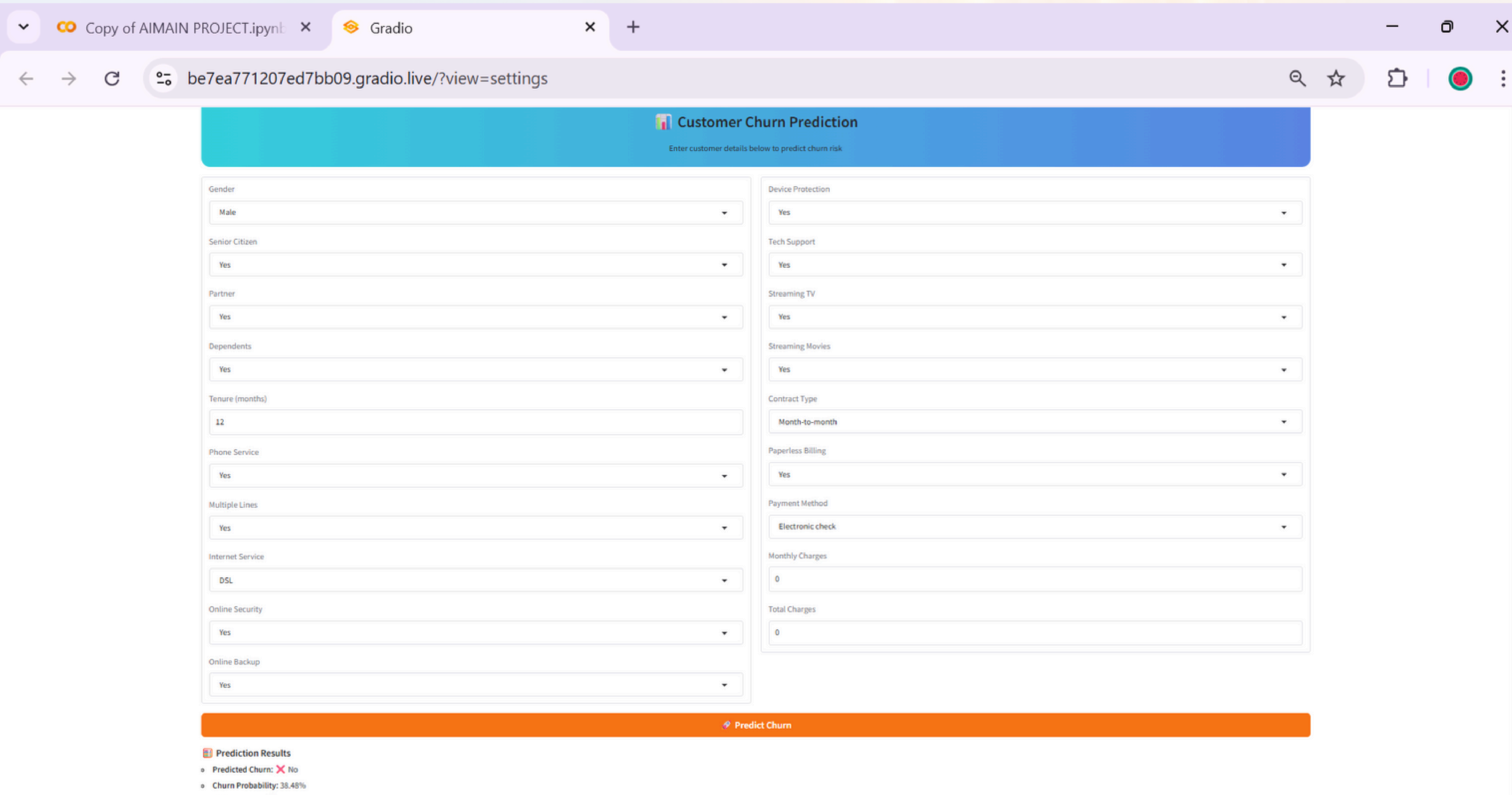
PREDICTED PROBABILITY

RISK SEGMENTATION

LOW RISK (<40%)
MEDIUM RISK (40–70%)
HIGH RISK (>70%)

RETENTION STRATEGIES

- To make predictions more useful for business, we segmented customers based on their churn probability.
- Customers with churn probability less than 40% were classified as Low Risk, between 40–70% as Medium Risk, and above 70% as High Risk.
- This segmentation allows the telecom company to prioritize retention strategies, focusing more on high-risk customers while maintaining engagement with medium- and low-risk groups.
- Our project showcased 77% accuracy.


Copy of AIMAIN PROJECT.ipynb
Gradio
be7ea771207ed7bb09.gradio.live/?view=settings
Customer Churn Prediction
Enter customer details below to predict churn risk
Gender
Male
Senior Citizen
Yes
Partner
Yes
Dependents
Yes
Tenure (months)
12
Phone Service
Yes
Multiple Lines
Yes
Internet Service
DSL
Online Security
Yes
Online Backup
Yes
Device Protection
Yes
Tech Support
Yes
Streaming TV
Yes
Streaming Movies
Yes
Contract Type
Month-to-month
Paperless Billing
Yes
Payment Method
Electronic check
Monthly Charges
0
Total Charges
0
Predict Churn
Prediction Results
Predicted Churn: ❌ No
Churn Probability: 38.48%
Risk Category: Low

BUSINESS IMPACT & FUTURE INNOVATION

CHURN PREDICTION BOT - 'STAYO"

- **Early Churn Prediction**

The bot uses a trained machine learning model to identify customers who are at high or medium risk of churn before they actually leave.

- **Personalized Retention Offers**

Based on the customer's churn probability, the bot automatically suggests or delivers personalized offers/discounts, loyalty rewards, or better plans to keep them engaged.

- **Automated Engagement**

Customers at risk receive proactive communication (e.g., through chat, email, or SMS) via the bot, making the process fast, consistent, and scalable.

- **Business Value**

By reducing churn and improving retention, the bot directly contributes to higher revenue, customer lifetime value (CLV), and overall profitability.

- **Continuous Learning**

The bot adapts over time by learning from new customer behavior and feedback, ensuring that predictions and offers remain accurate and effective.

CONCLUSION

- This project demonstrates how machine learning can help telecom companies reduce churn and improve customer loyalty.
- By predicting churn early, companies can proactively offer retention programs and reduce revenue loss.
- Our CatBoost model achieved high performance and identified tenure, contract type, and monthly charges as key drivers of churn.
- In the future, this system can be deployed into telecom CRM systems for real-time predictions and automated customer engagement.

THANK YOU!